

# B1 Algebra — Answer Key

IGCSE Mathematics 0580 | Full Solutions

## B1\_Q1: Algebraic Expressions — Problems

**F** Foundation — Answer:  $3x + 10y$

Group like terms:  $(5x - 2x) + (3y + 7y) = 3x + 10y$

**F** Foundation — Answer:  $8x + 12$

$4(2x + 3) = 4 \times 2x + 4 \times 3 = 8x + 12$

**F** Foundation — Answer:  $5a - 2b + 4$

Group a:  $7a - 2a = 5a$ . Group b:  $3b - 5b = -2b$ . Constant:  $+4$ . Result:  $5a - 2b + 4$

**F** Foundation — Answer:  $6x + 1$

$3(2x - 1) + 4 = 6x - 3 + 4 = 6x + 1$

**C** Core — Answer:  $12x - 7$

$2(3x + 4) = 6x + 8$ ,  $3(2x - 5) = 6x - 15$ . Sum:  $6x + 8 + 6x - 15 = 12x - 7$

**C** Core — Answer:  $x^2 + 10x + 21$

FOIL: First  $x \cdot x = x^2$ , Outer  $x \cdot 7 = 7x$ , Inner  $3 \cdot x = 3x$ , Last  $3 \cdot 7 = 21$ . Sum:  $x^2 + 10x + 21$

**C** Core — Answer:  $9x - 17$

$5(2x-1)=10x-5$ ,  $-3(x+4)=-3x-12$ . Full:  $10x-5-3x-12+2x = (10x-3x+2x)+(-5-12)=9x-17$

**C** Core — Answer:  $x^2 + 10x + 25$

$(x+5)^2 = x^2 + 2(x)(5) + 5^2 = x^2 + 10x + 25$

**H** Challenge — Answer:  $6x^2 - 5x - 4$

FOIL:  $(2x)(3x)=6x^2$ ,  $(2x)(-4)=-8x$ ,  $(1)(3x)=3x$ ,  $(1)(-4)=-4$ . Sum:  $6x^2 - 8x + 3x - 4 = 6x^2 - 5x - 4$

**H** Challenge — Answer:  $4x^2 - 9$

This is the difference of two squares:  $(2x+3)(2x-3) = (2x)^2 - 3^2 = 4x^2 - 9$

**H** Challenge — Answer:  $4x + 13$

$(x+2)^2 = x^2 + 4x + 4$ .  $(x-3)(x+3) = x^2 - 9$ . Subtract:  $x^2 + 4x + 4 - (x^2 - 9) = x^2 + 4x + 4 - x^2 + 9 = 4x + 13$

**H** Challenge — Answer:  $x^3 + 2x^2 - 5x + 12$

Multiply:  $x(x^2-2x+3) = x^3 - 2x^2 + 3x$ .  $4(x^2-2x+3) = 4x^2 - 8x + 12$ . Sum:  $x^3 + (-2x^2+4x^2) + (3x-8x) + 12 = x^3 + 2x^2 - 5x + 12$

**X** Extension — Answer:  $V = x^3 + 2x^2 - x - 2$ ,  $V(5) = 168$

Volume =  $(x+2)(x+1)(x-1)$ . First:  $(x+1)(x-1) = x^2-1$ . Then:  $(x+2)(x^2-1) = x(x^2-1) + 2(x^2-1) = x^3 - x + 2x^2 - 2 = x^3 + 2x^2 - x - 2$ . At  $x=5$ :  $125 + 50 - 5 - 2 = 168 \text{ cm}^3$

**X** Extension — Answer: Proof given

Let the odd numbers be  $2n+1$  and  $2n+3$ . Product:  $(2n+1)(2n+3) = 4n^2 + 6n + 2n + 3 = 4n^2 + 8n + 3 = 4(n^2 + 2n) + 3$ . This is  $4k + 3$  where  $k = n^2 + 2n$ , which is 3 more than a multiple of 4, i.e., one less than  $4k + 4 = 4(k+1)$ . So it is one less than a multiple of 4.

**X** Extension — Answer:  $2x + 6$

$\frac{x^2+4x+3}{x+1} = \frac{(x+1)(x+3)}{x+1} = x+3$  (for  $x \neq -1$ ).  $\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3$  (for  $x \neq 3$ ). Sum:  $(x+3)+(x+3)=2x+6$

**X** Extension — Answer:  $(2x+5)(3x-2)$  or  $(3x-2)(2x+5)$  or  $(6x-5)(x+2)$  etc.

We need  $ac=6$ ,  $ad+bc=11$ ,  $bd=-10$ . Factor pairs for 6:  $(1,6),(2,3),(3,2),(6,1),(-1,-6),(-2,-3),(-3,-2),(-6,-1)$ . Factor pairs for -10:  $(1,-10),(-1,10),(2,-5),(-2,5),(5,-2),(-5,2),(10,-1),(-10,1)$ . Trying  $(2x+5)(3x-2)$ : FOIL gives  $6x^2 - 4x + 15x - 10 = 6x^2 + 11x - 10$ . Works! So  $a=2, b=5, c=3, d=-2$  (and swapped  $a=3, b=-2, c=2, d=5$ ).

## B1\_Q2: Factorisation — Problems

**F** Foundation — Answer:  $3(2x + 3)$

HCF of  $6x$  and  $9$  is  $3$ .  $6x+9 = 3(2x+3)$

**F** Foundation — Answer:  $4x(3x - 2)$

HCF of  $12x^2$  and  $8x$  is  $4x$ .  $12x^2-8x = 4x(3x-2)$

**F** Foundation — Answer:  $(x + 4)(x - 4)$

Difference of two squares:  $x^2-16 = x^2-4^2 = (x+4)(x-4)$

**F** Foundation — Answer:  $(x + 2)(x + 3)$

Numbers with product 6 and sum 5: 2 and 3. So  $(x+2)(x+3)$

**C** Core — Answer:  $3(x + 3)(x - 3)$

First take out HCF:  $3(x^2-9)$ . Then difference of squares:  $3(x+3)(x-3)$

**C** Core — Answer:  $(x - 3)(x - 5)$

Numbers with product 15 and sum -8: -3 and -5. So  $(x-3)(x-5)$

**C** Core — Answer:  $(x - 4)(x + 3)$

Numbers with product -12 and sum -1: -4 and +3. So  $(x-4)(x+3)$

**C** Core — Answer:  $(3x + 2y)(3x - 2y)$

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x+2y)(3x-2y)$$

**H** Challenge — Answer:  $(2x + 1)(x + 2)$

$ac=2 \times 2=4$ . Numbers with product 4 and sum 5: 4 and 1. Rewrite:  $2x^2+4x+x+2 = 2x(x+2)+1(x+2) = (x+2)(2x+1)$

**H** Challenge — Answer:  $(3x + 2)(x - 4)$

$ac=3 \times (-8)=-24$ . Numbers with product -24 and sum -10: -12 and 2. Rewrite:  $3x^2-12x+2x-8 = 3x(x-4)+2(x-4) = (x-4)(3x+2)$

**H** Challenge — Answer:  $(2x - 3)(3x - 2)$

$ac=6 \times 6=36$ . Numbers with product 36 and sum -13: -9 and -4. Rewrite:  $6x^2-9x-4x+6 = 3x(2x-3)-2(2x-3) = (2x-3)(3x-2)$

**H** Challenge — Answer:  $(3x - 2)(2x + 3)$

$ac=6 \times (-6)=-36$ . Numbers with product -36 and sum 5: 9 and -4. Rewrite:  $6x^2+9x-4x-6 = 3x(2x+3)-2(2x+3) = (2x+3)(3x-2)$

**X** Extension — Answer:  $(x + 5)(x - 2)(x - 1)(x + 4)$

Let  $u = x^2 + 3x$ . Then expression  $= u^2 - 14u + 40 = (u-4)(u-10)$ . Sub back:  $(x^2+3x-4)(x^2+3x-10)$ . Factor each:  $(x+4)(x-1)(x+5)(x-2)$

**X** Extension — Answer:  $\frac{x-2}{x-4}$ ,  $x \neq -2, 4$

Numerator:  $x^2-4 = (x+2)(x-2)$ . Denominator:  $x^2-2x-8 = (x+2)(x-4)$ . Cancel  $(x+2)$ :  $\frac{x-2}{x-4}$ ,  $x \neq -2, 4$

**X** Extension — Answer: Proof given

$n^3 - n = n(n^2-1) = n(n-1)(n+1)$ . This is the product of three consecutive integers, so it always contains a multiple of 2 (at least one even number) and a multiple of 3 (one of every three consecutive numbers is divisible by 3). Therefore it is divisible by  $2 \times 3 = 6$ .

**X** Extension — Answer:  $(x - 2)(x + 2)(x - 3)$

Group:  $(x^3-3x^2) + (-4x+12) = x^2(x-3) - 4(x-3) = (x-3)(x^2-4) = (x-3)(x+2)(x-2)$

## B1\_Q3: Linear Equations — Problems

**F** Foundation — Answer:  $x = 5$

Subtract 7 from both sides:  $x = 12 - 7 = 5$

**F** Foundation — Answer:  $x = 6$

Divide both sides by 3:  $x = 18 / 3 = 6$

**F** Foundation — Answer:  $x = 4$

Subtract 5:  $2x = 8$ . Divide by 2:  $x = 4$

**F** Foundation — Answer:  $x > 10$

Add 3 to both sides:  $x > 10$

**C** Core — Answer:  $x = 4$

Subtract 2x:  $3x - 3 = 9$ . Add 3:  $3x = 12$ . Divide:  $x = 4$

**C** Core — Answer:  $x = 3$

Expand:  $6x - 2 = 16$ . Add 2:  $6x = 18$ . Divide:  $x = 3$

**C** Core — Answer:  $x \leq 6$

Add 7:  $3x \leq 18$ . Divide:  $x \leq 6$

**C** Core — Answer:  $x = -3$

Add 2x:  $7 = 5x + 22$ . Subtract 22:  $-15 = 5x$ . Divide:  $x = -3$

**H** Challenge — Answer:  $x = 5.5$

Expand:  $6x - 3 = 4x + 8$ . Subtract 4x:  $2x - 3 = 8$ . Add 3:  $2x = 11$ . Divide:  $x = 5.5$

**H** Challenge — Answer:  $x \leq 1$

Add 3x:  $12 \geq 8x + 4$ . Subtract 4:  $8 \geq 8x$ . Divide:  $1 \geq x$ , so  $x \leq 1$

**H** Challenge — Answer:  $x = 12$

Multiply by 6:  $3x + 2x = 60$ .  $5x = 60$ , so  $x = 12$

**H** Challenge — Answer:  $x = 14$

Cross-multiply:  $2(2x - 1) = 3(x + 4)$ .  $4x - 2 = 3x + 12$ .  $x = 14$

**X** Extension — Answer: Length = 10 cm, width = 7 cm

Let width =  $w$ , length =  $w+3$ . Perimeter:  $2(w + w+3) = 34$ .  $2(2w+3)=34$ ,  $4w+6=34$ ,  $4w=28$ ,  $w=7$ . Length =  $7+3=10$  cm

**X** Extension — Answer:  $-4 < x \leq 3$

Subtract 3 from all parts:  $-8 < 2x \leq 6$ . Divide by 2:  $-4 < x \leq 3$

**X** Extension — Answer:  $x = 0$

Multiply by LCM 12:  $4(2x+1) - 3(x-2) = 10$ .  $8x+4-3x+6=10$ .  $5x+10=10$ .  $5x=0$ .  $x=0$ . Check:  $\frac{0+1}{3} - \frac{0-2}{4} = \frac{1}{3} - \frac{-2}{4} = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$  ✓

**X** Extension — Answer: Mary is 36 years old

Let son's age =  $x$ . Mary's age =  $3x$ . In 12 years: Mary  $3x+12$ , son  $x+12$ . Equation:  $3x+12 = 2(x+12)$ .  $3x+12 = 2x+24$ .  $x=12$ . So son is 12, Mary is  $3(12)=36$

## B1\_Q4: Simultaneous Equations — Problems

**F** Foundation — Answer:  $x = 5$ ,  $y = 2$

Add equations:  $2x = 10$ ,  $x=5$ . Substitute:  $5+y=7$ ,  $y=2$

**F** Foundation — Answer:  $x = 6$ ,  $y = 4$

Subtract eq1 from eq2:  $(2x+y)-(x+y)=16-10$ ,  $x=6$ . Substitute:  $6+y=10$ ,  $y=4$

**F** Foundation — Answer:  $x = 3$ ,  $y = 6$

Substitute:  $x+2x=9$ ,  $3x=9$ ,  $x=3$ . Then  $y=2(3)=6$

**F** Foundation — Answer:  $x = 5$ ,  $y = 2$

Add:  $3x=15$ ,  $x=5$ . Substitute:  $2(5)+y=12$ ,  $10+y=12$ ,  $y=2$

**C** Core — Answer:  $x = 4$ ,  $y = 1$

Subtract:  $(5x+2y)-(3x+2y)=22-14$ ,  $2x=8$ ,  $x=4$ . Substitute:  $3(4)+2y=14$ ,  $12+2y=14$ ,  $y=1$

**C** Core — Answer:  $x = 3$ ,  $y = 2$

Multiply eq1 by 3:  $6x+9y=36$ . Eq2 by 2:  $6x+4y=26$ . Subtract:  $5y=10$ ,  $y=2$ . Substitute:  $2x+6=12$ ,  $2x=6$ ,  $x=3$

**C** Core — Answer:  $x = 2$ ,  $y = 5$

Substitute:  $2x+5(3x-1)=29$ ,  $2x+15x-5=29$ ,  $17x=34$ ,  $x=2$ . Then  $y=3(2)-1=5$

**C** Core — Answer:  $x = 5$ ,  $y = 1$

Add:  $7x = 35$ ,  $x=5$ . Substitute:  $2(5)-3y=7$ ,  $10-3y=7$ ,  $-3y=-3$ ,  $y=1$

**H** Challenge — Answer:  $x = 3$ ,  $y = 1$

Multiply eq1 by 2:  $6x-8y=10$ . Eq2 by 3:  $6x+15y=33$ . Subtract:  $-23y=-23$ ,  $y=1$ . Substitute:  $3x-4=5$ ,  $3x=9$ ,  $x=3$

**H** Challenge — Answer:  $x = 6$ ,  $y = 2$

Multiply eq1 by 2:  $x+2y=10$ . Eq2 by 2:  $2x-y=10$ . From eq1':  $x=10-2y$ . Substitute into eq2':  $2(10-2y)-y=10$ ,  $20-4y-y=10$ ,  $20-5y=10$ ,  $5y=10$ ,  $y=2$ . Then  $x=10-2(2)=6$

**H** Challenge — Answer: 20 and 8

Let numbers be  $x$  and  $y$  with  $x>y$ .  $x+y=28$ ,  $x-y=12$ . Add:  $2x=40$ ,  $x=20$ . Then  $20+y=28$ ,  $y=8$

**H** Challenge — Answer:  $x = 1$ ,  $y = 2$

Multiply eq1 by 2:  $8x+6y=20$ . Eq2 by 3:  $9x-6y=-3$ . Add:  $17x=17$ ,  $x=1$ . Substitute:  $4(1)+3y=10$ ,  $4+3y=10$ ,  $3y=6$ ,  $y=2$

**X** Extension — Answer: Apple = \$0.50, Banana = \$0.40

Let  $a$  = cost of apple,  $b$  = cost of banana.  $3a+4b=3.10$ ,  $5a+2b=3.30$ . Multiply eq2 by 2:  $10a+4b=6.60$ . Subtract eq1:  $7a=3.50$ ,  $a=0.50$ . Substitute:  $3(0.50)+4b=3.10$ ,  $1.50+4b=3.10$ ,  $4b=1.60$ ,  $b=0.40$

**X** Extension — Answer:  $x = 1$ ,  $y = 2$ ,  $z = 3$

Add eq1 and eq2:  $2x+2z=8$ ,  $x+z=4$ . Add eq1 and eq3:  $2x+2y=6$ ,  $x+y=3$ . Subtract eq2 from eq1:  $2y=4$ ,  $y=2$ . Then  $x+2=3$ ,  $x=1$ . Then  $1+z=4$ ,  $z=3$

**X** Extension — Answer:  $k = 14$

Infinitely many solutions occur when one equation is a multiple of the other. Eq2 is 2 times eq1:  $2(2x+3y)=2(7)$ , so  $4x+6y=14$ . Therefore  $k=14$ .

**X** Extension — Answer:  $x = 2$ ,  $y = 4$  or  $x = -3$ ,  $y = 9$

Substitute:  $x+x^2=6$ ,  $x^2+x-6=0$ ,  $(x+3)(x-2)=0$ ,  $x=-3$  or  $x=2$ . When  $x=-3$ ,  $y=9$ . When  $x=2$ ,  $y=4$

## B1\_Q5: Algebraic Fractions — Problems

**F** Foundation — Answer:  $\frac{x}{2}$

Cancel common factor 3:  $\frac{3x}{6} = \frac{x}{2}$

**F** Foundation — Answer:  $2x$

$\frac{4x^2}{2x} = \frac{4}{2} \times \frac{x^2}{x} = 2x$

**F** Foundation — Answer: \$4\$

$$\frac{2x \times 6}{3 \times x} = \frac{12x}{3x} = 4$$

**F** Foundation — Answer: \$2\$

$$\frac{x}{2} \times \frac{4}{x} = \frac{4x}{2x} = 2$$

**C** Core — Answer: \$x - 3\$, \$x \neq -3\$

$$x^2 - 9 = (x+3)(x-3). \text{ Cancel } (x+3): \frac{(x+3)(x-3)}{x+3} = x-3$$

**C** Core — Answer: \$\frac{5}{x}\$

$$\text{Common denominator } x: \frac{3+2}{x} = \frac{5}{x}$$

**C** Core — Answer: \$\frac{3}{x+1}\$

$$\text{Common denominator } x+1: \frac{5-2}{x+1} = \frac{3}{x+1}$$

**C** Core — Answer: \$\frac{x+3}{2}\$, \$x \neq 0\$

$$\text{Factorise numerator: } \frac{2x(x+3)}{4x} = \frac{x+3}{2}$$

**H** Challenge — Answer: \$\frac{5x+2}{x(x+1)}\$

$$\text{Common denominator } x(x+1): \frac{2(x+1)}{x(x+1)} + \frac{3x}{x(x+1)} = \frac{2x+2+3x}{x(x+1)} = \frac{5x+2}{x(x+1)}$$

**H** Challenge — Answer: \$\frac{x-3}{x-2}\$, \$x \neq \pm 2\$

$$\text{Numerator: } x^2 - x - 6 = (x-3)(x+2). \text{ Denominator: } x^2 - 4 = (x+2)(x-2). \text{ Cancel } (x+2)$$

**H** Challenge — Answer: \$\frac{x+8}{(x-1)(x+2)}\$

$$\text{Common denominator } (x-1)(x+2): \frac{3(x+2)-2(x-1)}{(x-1)(x+2)} = \frac{3x+6-2x+2}{(x-1)(x+2)} = \frac{x+8}{(x-1)(x+2)}$$

**H** Challenge — Answer: \$x = -\frac{1}{4}\$

$$\text{Multiply by } 2x: 6 + 4x = 5. 4x = -1, x = -\frac{1}{4}$$

**X** Extension — Answer: \$-\frac{1}{x}\$

$$\text{Numerator: } \frac{(x-1)-(x+1)}{(x+1)(x-1)} = \frac{-2}{(x+1)(x-1)}. \text{ Denominator: } \frac{(x-1)+(x+1)}{(x+1)(x-1)} = \frac{2x}{(x+1)(x-1)}. \text{ Divide: } \frac{-2}{(x+1)(x-1)} \times \frac{(x+1)(x-1)}{2x} = -\frac{1}{x}$$

**X** Extension — Answer: \$x = 2\$

$$\text{Multiply by } (x+1)(x-1): (2x-1)(x-1)-(x-2)(x+1)=(x+1)(x-1). \text{ Expand: } (2x^2-3x+1)-(x^2-x-2)=x^2-1. \text{ Simplify: } x^2-2x+3=x^2-1, -2x=-4, x=2$$

**X** Extension — Answer:  $\frac{x^2+5}{(x+2)(x-1)}$

Common denominator  $(x+2)(x-1)$ :  $\frac{(2x+1)(x-1)-(x-3)(x+2)}{(x+2)(x-1)}$ . Numerator:  $(2x^2-x-1)-(x^2-x-6) = x^2+5$

**X** Extension — Answer:  $x = 3$  hours

Rates: A fills  $\frac{1}{x}$  per hour, B fills  $\frac{1}{x+3}$  per hour. Combined:  $\frac{1}{x} + \frac{1}{x+3} = \frac{1}{2}$ . Multiply by  $2x(x+3)$ :  $2(x+3)+2x=x(x+3)$ ,  $4x+6=x^2+3x$ ,  $x^2-x-6=0$ ,  $(x-3)(x+2)=0$ ,  $x=3$  or  $x=-2$  (reject). So  $x=3$  hours.

## B1\_Q6: Sequences — Problems

**F** Foundation — Answer: \$14, 17\$

Common difference  $d=3$ . Next:  $11+3=14$ ,  $14+3=17$

**F** Foundation — Answer:  $3n + 4$

$d=3$ , so  $u_n = 3n + b$ . At  $n=1$ :  $3(1)+b=7$ ,  $b=4$ . So  $u_n = 3n+4$

**F** Foundation — Answer: \$48\$

$u_{10} = 5(10)-2 = 50-2 = 48$

**F** Foundation — Answer: \$3, 7, 11, 15, 19\$

Start at 3, add 4 each time: 3, 7, 11, 15, 19

**C** Core — Answer: 15th term

$4n-3=57$ ,  $4n=60$ ,  $n=15$ . So the 15th term is 57

**C** Core — Answer:  $4n - 5$

$d=4$ ,  $u_n=4n+b$ . At  $n=1$ :  $4+b=-1$ ,  $b=-5$ . So  $u_n=4n-5$

**C** Core — Answer: \$36\$

First differences: 3,5,7,9. Second difference: 2 (constant). Next first diff: 11. Next term:  $25+11=36$

**C** Core — Answer:  $15 - 4n$

$d=-4$ ,  $u_n=-4n+b$ . At  $n=1$ :  $-4+b=11$ ,  $b=15$ . So  $u_n=15-4n$

**H** Challenge — Answer:  $n^2 + n$

First diffs: 4,6,8,10. Second diff: 2 (constant).  $a=2/2=1$ . Subtract  $n^2$ : 1,2,3,4,5 which is  $n$ . So  $u_n=n^2+n$

**H** Challenge — Answer:  $2n^2 + n + 2$

First diffs: 7,11,15,19. Second diff: 4.  $a=4/2=2$ . Subtract  $2n^2$ : 3,4,5,6,7 which is  $n+2$ . So  $u_n=2n^2+n+2$

**H** Challenge — Answer:  $n = 6$

$n^2+3n=54$ ,  $n^2+3n-54=0$ ,  $(n+9)(n-6)=0$ ,  $n=6$  or  $n=-9$  (reject). So  $n=6$

**H** Challenge — Answer:  $u_1 = 2$ ,  $u_5 = 86$

$u_1 = 4-3+1=2$ .  $u_5 = 4(25)-3(5)+1 = 100-15+1 = 86$

**X** Extension — Answer:  $u_n = 4n + 1$

$u_n = S_n - S_{n-1}$ .  $S_{n-1}=2(n-1)^2+3(n-1)=2(n^2-2n+1)+3n-3=2n^2-4n+2+3n-3=2n^2-n-1$ . So  $u_n = (2n^2+3n)-(2n^2-n-1)=4n+1$

**X** Extension — Answer:  $a=3$ ,  $b=-2$ ,  $c=1$

$n=1$ :  $a+b+c=2$ .  $n=2$ :  $4a+2b+c=9$ .  $n=3$ :  $9a+3b+c=22$ . Subtract eq1 from eq2:  $3a+b=7$ . Subtract eq2 from eq3:  $5a+b=13$ . Subtract:  $2a=6$ ,  $a=3$ . Then  $9+b=7$ ,  $b=-2$ . Then  $3-2+c=2$ ,  $c=1$ . Check:  $u_2=3(4)-2(2)+1=9$  ✓,  $u_3=27-6+1=22$  ✓. So  $u_n=3n^2-2n+1$

**X** Extension — Answer: Proof given

$n^2+3n+1 = n(n+3)+1$ . If  $n$  is even,  $n(n+3)$  is even, so  $n(n+3)+1$  is odd. If  $n$  is odd,  $n+3$  is even, so  $n(n+3)$  is even, so  $n(n+3)+1$  is odd. In both cases the result is odd.

**X** Extension — Answer:  $M_n = 3n+1$ ,  $n = 20$  squares

Sequence: 4,7,10,...  $d=3$ , so  $M_n=3n+1$ . Set  $3n+1=61$ ,  $3n=60$ ,  $n=20$

## B1\_Q7: Functions and Graphs — Problems

**F** Foundation — Answer:  $11$

$f(3) = 2(3) + 5 = 6 + 5 = 11$

**F** Foundation — Answer:  $g(3) = 5$ ,  $g(0) = -4$

$g(3) = 3^2 - 4 = 9 - 4 = 5$ .  $g(0) = 0 - 4 = -4$

**F** Foundation — Answer:  $3$

In  $y = mx + c$ ,  $m$  is the gradient. So  $m = 3$

**F** Foundation — Answer:  $7$

In  $y = mx + c$ ,  $c$  is the y-intercept. So  $c = 7$

**C** Core — Answer: \$2\$

$$m = \frac{9-3}{4-1} = \frac{6}{3} = 2$$

**C** Core — Answer: \$y = 3x - 1\$

$$y - 5 = 3(x - 2), y - 5 = 3x - 6, y = 3x - 1$$

**C** Core — Answer: \$x = 5\$

$$3x - 2 = 13, 3x = 15, x = 5$$

**C** Core — Answer: \$y = 2x - 3\$

$$m = \frac{5-(-3)}{4-0} = \frac{8}{4} = 2. \text{ The y-intercept is } -3 \text{ (since } (0, -3)). \text{ So } y = 2x - 3$$

**H** Challenge — Answer: \$-\frac{1}{2}\$

$$\text{Perpendicular gradients multiply to } -1. 2 \times m = -1, \text{ so } m = -\frac{1}{2}$$

**H** Challenge — Answer: \$y = \frac{1}{3}x + \frac{1}{3}\$

$$\text{Gradient of given line: } -3. \text{ Perpendicular gradient: } \frac{1}{3} \text{ (since } -3 \times \frac{1}{3} = -1). \text{ Equation: } y-1 = \frac{1}{3}(x-2), y = \frac{1}{3}x - \frac{2}{3} + 1 = \frac{1}{3}x + \frac{1}{3}$$

**H** Challenge — Answer: \$y = 4x + 5\$

$$\text{Parallel lines have same gradient: } m=4. \text{ Line passes through } (0,5) \text{ so y-intercept } c=5. \text{ Equation: } y=4x+5$$

**H** Challenge — Answer: \$f(g(3)) = 17\$, \$g(f(3)) = 25\$

$$g(3)=3^2=9, \text{ so } f(g(3)) = f(9) = 2(9)-1=17. f(3)=2(3)-1=5, \text{ so } g(f(3))=g(5)=5^2=25$$

**X** Extension — Answer: \$\{2, 3, 6\}\$

$$f(-2)=4+2=6, f(-1)=1+2=3, f(0)=0+2=2, f(1)=1+2=3, f(2)=4+2=6. \text{ Range: } \{2,3,6\}$$

**X** Extension — Answer: \$f^{-1}(x) = \frac{x+3}{x-2}\$, \$x \neq 2\$

$$\text{Let } y = \frac{2x+3}{x-1}. \text{ Swap } x \text{ and } y: x = \frac{2y+3}{y-1}. \text{ Multiply: } x(y-1)=2y+3, xy-x=2y+3, xy-2y=x+3, y(x-2)=x+3, y = \frac{x+3}{x-2}. \text{ So } f^{-1}(x) = \frac{x+3}{x-2}, x \neq 2$$

**X** Extension — Answer: \$(2, 4)\$

$$\text{Substitute } y=3x-2 \text{ into } 2x+y=8: 2x+3x-2=8, 5x=10, x=2. \text{ Then } y=3(2)-2=4. \text{ Intersection: } (2,4)$$

**X** Extension — Answer: \$f(x)=2x+3\$. 12 km costs \$27. For \$25, you can go 11 km.

$$f(x)=2x+3. \text{ For } x=12: f(12)=24+3=27 \text{ dollars. For } 25: 2x+3=25, 2x=22, x=11 \text{ km}$$